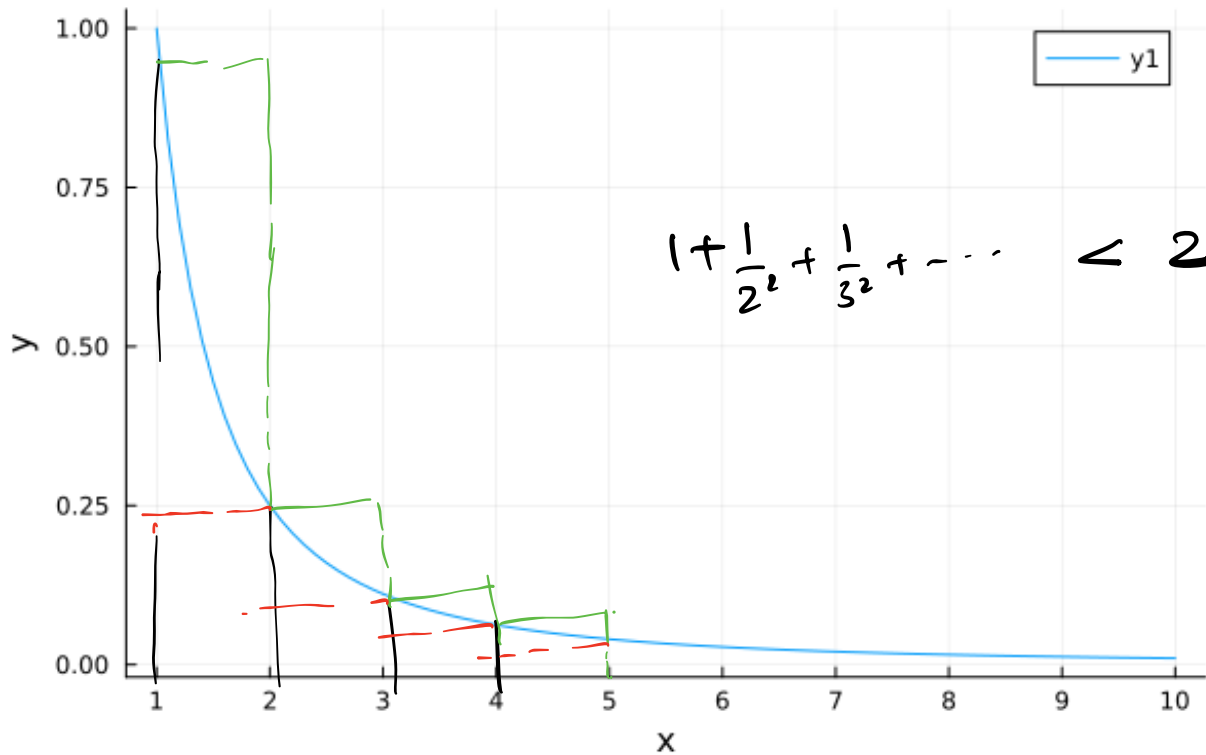


Plot of $y = 1/x^2$



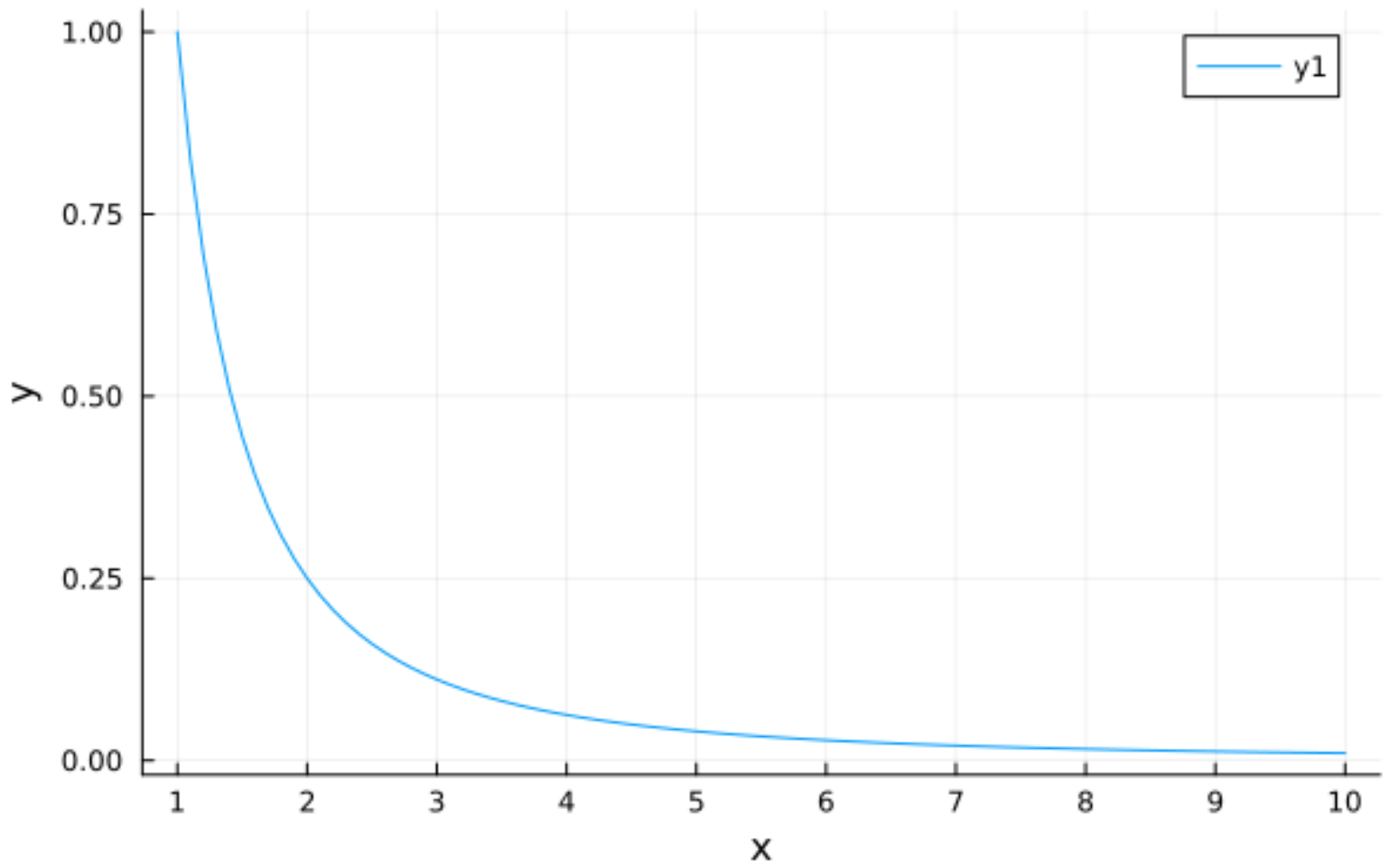
Green: $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

Red: $\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

$$\underbrace{\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots}_{\text{Red}} \leq \int_1^{\infty} \frac{1}{x^2} dx \leq \underbrace{1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots}_{\text{Green}}$$

$S_n = \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{(n+1)^2}$ is monotonically increase and bounded above.

Plot of $y = 1/x^2$



Series

$$\sum_{k=1}^{\infty} a_k$$

$$\text{Let } S_n = \sum_{k=1}^n a_k$$

If $\lim (S_n)$ exists, then we say that the series $\sum_{k=1}^{\infty} a_k$ converges. else

we say it diverges.

S_n : partial sums

$$R_n: \sum_{k=1}^{\infty} a_k - S_n = \sum_{n+1}^{\infty} a_k$$

a 's called the remainder.

Tool 7: Geometric series

$$a_{n+1} = d a_n$$

$$a_1 = 1$$

$$\sum a_k = \underbrace{1 + d + d^2 + d^3 + \dots}$$

$$S_n = 1 + d + d^2 + \dots + d^{n-1} = \frac{d^n - 1}{d - 1} \quad d \neq 1$$

For $\sum a_k$ to exist, d^n must have a limit as $n \rightarrow \infty$

$$-1 < d < 1$$

the series $1 + d + d^2 + \dots$ exists/converges if & only if

$$|d| < 1.$$

$$\left\{ \begin{array}{l} a_{n+1} = \sqrt{a_n a_{n-1}} \\ a_{n+1} = \frac{a_n + a_{n-1}}{2} \\ \underbrace{a_{n+1} - a_n}_{b_{n+1}} \\ = - \left(\frac{a_n - a_{n-1}}{2} \right) \end{array} \right.$$

$$\frac{1}{1-d} = 1 + d + d^2 + d^3 + \dots - \frac{1-d}{1-d}$$

$$\frac{1}{2} = 1 - 1 + 1 - 1 + 1 - 1 + \dots$$

$$0 = 0 + 0 + 0 + \dots$$

$$= (1-1) + (1-1) + (1-1) + \dots$$

$$= \underbrace{1-1 + 1-1 + 1-1 + \dots}$$

$$= 1 + (-1+1) + (-1+1) + (-1+1) + \dots$$

$$= 1$$

Exercise: Let a_n be a sequence that satisfies

$$a_{n+1} = \frac{1}{2}(a_n + a_{n-1}) \quad n \geq 2.$$

Find an expression for a_n in terms of a_1 & a_2

Hint: $b_n = a_n - a_{n-1}$

For $1 + d + d^2 + \dots$

$$\text{the } R_n = \frac{1}{1-d} - \frac{d^n - 1}{d - 1} = \underbrace{\frac{d^n}{1-d}}$$

write

$0.327272727\dots$ as fraction.

$$= 0.3 + 0.027 + 0.00027 + \dots$$

$$= 0.3 + \frac{27}{10^3} + \frac{27}{10^5} + \frac{27}{10^7} + \dots$$

Sum by Geometric series e

$$\sum_{n=0}^{\infty} \left(\frac{1}{e}\right)^n = \frac{e}{e-1}$$

Harmonic Series:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}.$$

n^{th} Harmonic number

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \dots + \frac{1}{16}\right) + \dots$$

$$> 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{4} + \frac{1}{4}\right)} + \underbrace{\left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)}$$

$$+ \underbrace{\left(\frac{1}{16} + \frac{1}{16} + \dots\right)}$$

$$H_{2^n} > 1 + \frac{n}{2}$$

$$H_n > \log(n)$$

Can be shown that $H_n = \ln n + \text{const} + \underbrace{o(1)}_{\substack{\rightarrow \text{terms that} \\ \text{becomes smaller as} \\ n \rightarrow \infty}}$ $H_n = O(\ln n)$

$a_n \rightarrow 0$ as $n \rightarrow \infty$
 $\underbrace{\quad}_{o(1)}$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots$$

Tool 8: Divergence test.

If $\sum_{k=1}^{\infty} a_k$ converges, then $a_n \rightarrow 0$ as $n \rightarrow \infty$.

$$\sum_{k=1}^{\infty} \underbrace{\frac{k^2}{5k^2 + 19}}_{a_k}$$

diverges to ∞ .

$$S_n = \sum_{k=1}^n a_k$$

$$\underbrace{S_{n+1} - S_n}_{=} = \underbrace{a_{n+1}}_{=}$$

Take limits on both sides,

$$0 = \lim (S_{n+1} - S_n) = \lim a_{n+1}$$

$$\begin{aligned} \lim_{k \rightarrow \infty} a_k &= \lim_{k \rightarrow \infty} \frac{k^2}{5k^2 + 19} \\ &= \lim_{k \rightarrow \infty} \frac{1}{5 + 19/k^2} \\ &= \frac{1}{5} \end{aligned}$$

Tool 9: Integral test

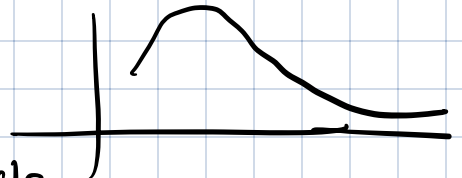
Suppose $a_n = f(n)$ for some function

$f: [1, \infty) \rightarrow \mathbb{R}$ that is continuous, monotonic decreasing and positive.

$$a_n = \frac{1}{n^2}$$

$$f(x) = \frac{1}{x^2}$$

$\sum_{k=1}^{\infty} a_k$ converges if & only if $\int_1^{\infty} f(x) dx$ exists.



① $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

check $\int_1^{\infty} \frac{1}{x^2} dx = 1$

② $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

$H_n \leq \ln n + \text{const.}$

$\int_1^{\infty} \frac{1}{x} dx = \ln(\infty)$

$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{4}} + \dots$

③ $1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \dots$

∞

$$\sum_{k=1}^{\infty} \frac{1}{k^p}$$

Case 1: $p \leq 0$ the series diverge by divergence test
 $1 + 2^{-p} + 3^{-p} + \dots$

Case 2: $0 < p \leq 1$
 diverges

$\int_1^{\infty} \frac{1}{x^p} dx$ exists only when $p > 1$.

Case 3: $p > 1$
 converges.

Exam: Approximate the sum

$$\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \frac{1}{(n+3)^2} + \dots$$

by integrals.

How is this related to
 $\int_{n+1}^{\infty} \frac{1}{x^2} dx$

Once convergence is established, the limit can be approximated by S_n for n large enough

$$S_n = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{\pi^2}{6}$$

$$S \approx S_5$$

$$Error = S - S_5 = \underbrace{\frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \dots}_{\text{Integral test}}$$

$$\int_6^{\infty} \frac{1}{x^2} dx \leq \int_5^{\infty} \frac{1}{x^2} dx$$

(Integral test)

Exercise: Find $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$ upto 4 decimal places,

$$S \approx S_n$$

the error in approximating S with S_n is at most $\int_n^{\infty} \frac{1}{x^2} dx$

$$\int_{n+1}^{\infty} \frac{1}{x^2} dx \leq |S - S_n| \leq \int_n^{\infty} \frac{1}{x^2} dx$$

pick n such that $\int_n^{\infty} \frac{1}{x^2} dx < 10^{-5}$.

Tool 10: Comparison test.

$$\sum_{n=0}^{\infty} \frac{1}{2^n + 1} < \sum_{n=0}^{\infty} \frac{1}{2^n} = 2$$

$$A = \sum_{n=1}^{\infty} \frac{1}{2^n - 1}$$

$\sum_{n=1}^{\infty} \frac{1}{2^n}$

$$B = \sum_{n=1}^{\infty} \frac{1}{2^n}$$

$$\frac{\frac{1}{2^n - 1}}{\frac{1}{2^n}} = \frac{1}{1 - \frac{1}{2^n}}$$

$n=2$ onwards, the n^{th} term of A is almost twice the

n^{th} term of B.

$n=3$ onwards, the n^{th} term of A is almost $\frac{8}{7}$ times the

n^{th} term of B.

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \underbrace{\sum_{n=3}^{\infty} a_n}_{\leq a_1 + a_2 + \underbrace{\sum_{n=3}^{\infty} b_n}_{\leq \frac{8}{7} \sum_{n=3}^{\infty} b_n}}$$

$\sum a_n, \sum b_n$ be series with positive terms.

if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$ then if $\sum a_n$ converges,

so does $\sum b_n$. if $\sum a_n$ diverges, so does $\sum b_n$.

Example:

$$\sum_{n=1}^{\infty} \underbrace{\frac{4n^4 + 3n^3 + 10n^2 + 1}{\sqrt{n^{10} + 1}}}_{b_n}$$

$$\sum \underbrace{\frac{1}{n}}_{a_n}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n^{10} + 1}}{4n^5 + 3n^4 + 10n^3 + n} = \underline{\underline{1/4}}$$

Tool 11: Alternating series test

Let $\sum_{n=1}^{\infty} (-1)^n b_n$ be an alternating series where $b_n \geq 0$ such that $0 \leq b_{n+1} \leq b_n$ and

$$\lim_{n \rightarrow \infty} b_n = 0.$$

then $\sum_{n=1}^{\infty} (-1)^n b_n$ converges.

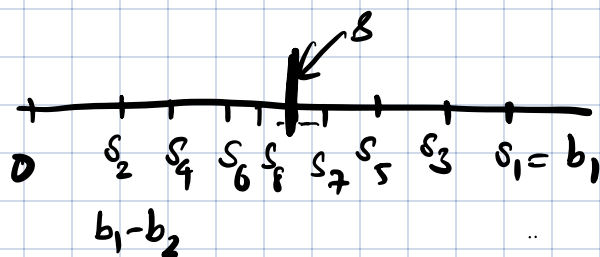
$$\frac{1}{e} = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \frac{1}{5!} - \dots = \sum_{n=1}^{\infty} (-1)^n b_n$$

$$b_n = \frac{1}{n!}$$

$$s_1 = b_1$$

$$s_2 = b_1 - b_2$$

$$s_3 = b_1 - b_2 + b_3$$



$$s_{101} - s_{100} = \underbrace{b_{101}}$$

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

s is between

s_{even} & s_{odd}

$$\boxed{|s - s_n| \leq b_{n+1}}$$

$$s_{2n} \leq s \leq s_{2n+1}$$

$$b_{2n+1}$$

Tool 12: Ratio test

Let $\sum a_n$ be a series such that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

(i) If $L > 1$, the series $\sum |a_n|$ diverges

(ii) If $L < 1$, the series $\sum |a_n|$ converges

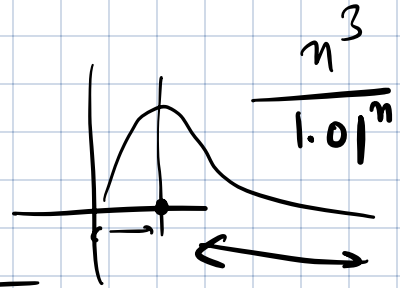
(iii) If $L = 1$, test inconclusive.

$$\underbrace{\sum a_n} \leq \underbrace{\sum |a_n|}$$

Example: $\sum_{n=1}^{\infty} \frac{n^3}{3^n}$

$$\frac{\frac{(n+1)^3}{3^{n+1}}}{\frac{n^3}{3^n}} = \left(\frac{n+1}{n}\right)^3 \frac{1}{3} \rightarrow \frac{1}{3}$$

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$$



Property:

If $\sum |a_n|$ converges, so does $\sum a_n$.

Power series:

$$a_0 + a_1x + a_2x^2 + a_3x^3 + \dots = \dots$$

$$a_0 + a_1 + a_2 + \dots$$

$$S_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

Example: $1 + x + x^2 + x^3 + \dots$ converges if &

$$S_n(x) = \frac{1-x^{n+1}}{1-x}$$

only if $|x| < 1$

$$\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$$

Apply ratio test

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(x-3)^{n+1}}{n+1}}{\frac{(x-3)^n}{n}} \right| = |x-3|$$

If $|x-3| < 1$ then the series converges.

$J_0(x)$

$$|| \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{2^{2n} (n!)^2} \quad \swarrow a_n$$

converges for all x .

$$\frac{4 \cdot \frac{x^{2n+2}}{2^{2n+2} ((n+1)!)^2}}{\frac{x^{2n}}{2^{2n} (n!)^2}} = \frac{x^2}{4(n+1)^2}$$

$\rightarrow 0$

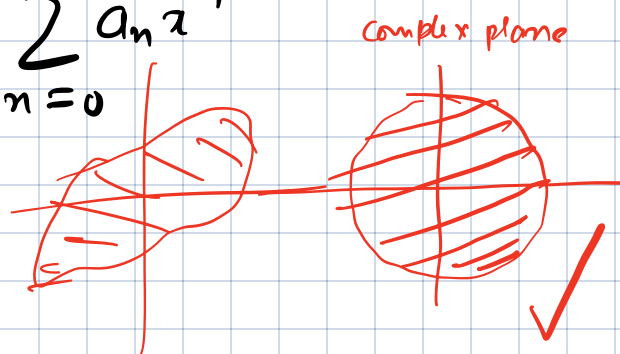
$L=0 < 1$
for all x .

Tool 13(x) Power series of the form $\sum_{n=0}^{\infty} a_n x^n$

$$\sum_{n=0}^{\infty} a_n x^n$$

has three possibilities

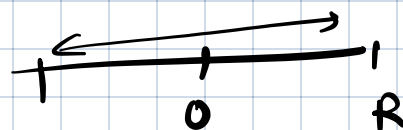
$R = \infty$ (i) Converges for all x .



(ii) there exists R such that the series
converges for $|x| < R$, diverges $|x| > R$.

$R = 0$ (iii) diverges for all $x \neq 0$.

R : radius of convergence
(Roc)



$$\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)}$$

$$\lim \left| \frac{x^{n+1}}{(n+1)(n+2)} \right| = |x|$$

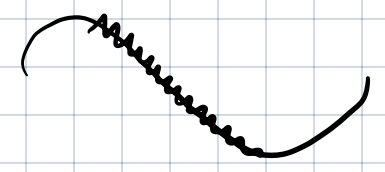
$$\left| \frac{x^n}{n(n+1)} \right|$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \lim \left| \frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}} \right| = \lim \left| \frac{x}{n+1} \right| = 0$$

Suppose a power series converges to $f(x)$ for $|x| < R$.

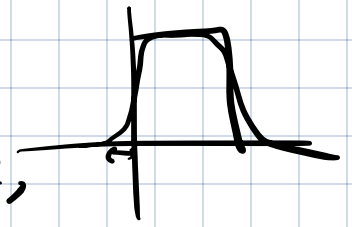
$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$



Tool 13 (b)

within the region of convergence, we can differentiate / integrate term by term.



$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Tool 14 (a) Taylor Series

Suppose f is a function s.t.

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \quad \text{for } |x| < \underline{R}. \quad (R > 0)$$

$$\text{then } a_n = \left. \frac{f^{(n)}(x)}{n!} \right|_{x=0}$$

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots$$

$$f(x) = \underbrace{f(0) + f'(0)x + \frac{f''(0)x^2}{2} + \dots}_{\nearrow 10}$$

Tool 14(b): Error in Taylor series

$$\text{Let } R_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} x^k$$

$$\text{then } |R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} M \quad \text{where } M = \max_{|x| < R} f^{(n+1)}(x)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

How many terms do I need to estimate e^5 to 1st decimal place?

pick n such that

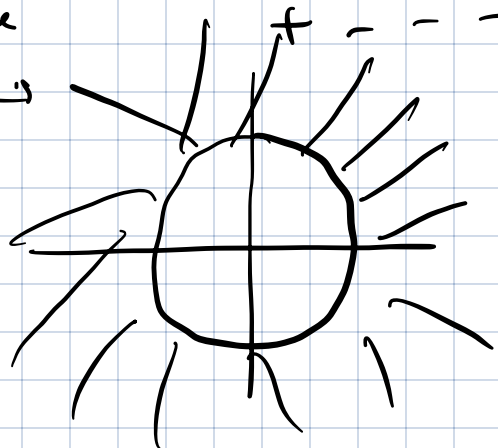
$$\frac{5^{n+1} e^5}{(n+1)!} < 10^{-2}$$

Recap: Power series

$$\sum_{n=0}^{\infty} a_n x^n$$

$$a_0 + a_1 x^{-1} + a_2 x^{-2} + a_3 x^{-3} + \dots + a_n x^{-n}$$

Region of convergence outside disc



$$\sum a_n x^n + \sum b_n x^n$$

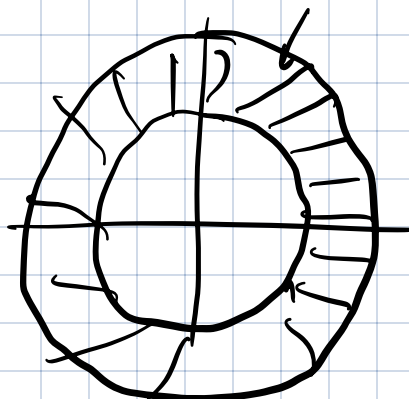
$S_1 \quad S_2$

the region of convergence $\supset S_1 + S_2$

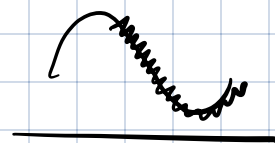
includes region of cgs $\supset S_1 \cap$ region of cgs of S_2

$$\underbrace{a_{-2} x^{-2} + a_{-1} x^{-1}}_{R_2} + a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

$\sum_{n=-\infty}^{\infty} a_n x^n$ R_1



$R_1 > R_2$



if $f(x) = \sum_{n=0}^{\infty} a_n x^n$, then $a_n = \frac{f^{(n)}(x)}{n!} \Big|_{x=0}$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Taylor series about $x=0$.

$$f(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n \quad \text{Taylor series about } x=x_0.$$

$$a_n = \frac{f^{(n)}(x)}{n!} \Big|_{x=x_0}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad \text{--- ①}$$

Find Taylor series for $\sin x$, $\cos x$ either by directly finding n^{th} derivatives at origin or using Euler's identity. --- ①.

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$(1+x)^k = 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots$$

k positive integer $\rightarrow$ Binomial theorem.

Exer: Find Taylor series of e^{-x^2}

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{6} + \dots \quad x \rightarrow -x^2$$

$$\int_0^{\infty} e^{-x^2} dx$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

$$\int_0^1 e^{-x^2} dx = 1 - \frac{1}{3} + \frac{1}{5 \times 2!} - \frac{1}{7 \times 6} + \frac{1}{9 \times 4!} - \dots$$

$$\int_1^2 \frac{e^{-x^2}}{x} dx$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\operatorname{sech} x = \frac{2}{e^x + e^{-x}}$$

$$\begin{cases} \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots \\ \tan^{-1} x = \end{cases}$$

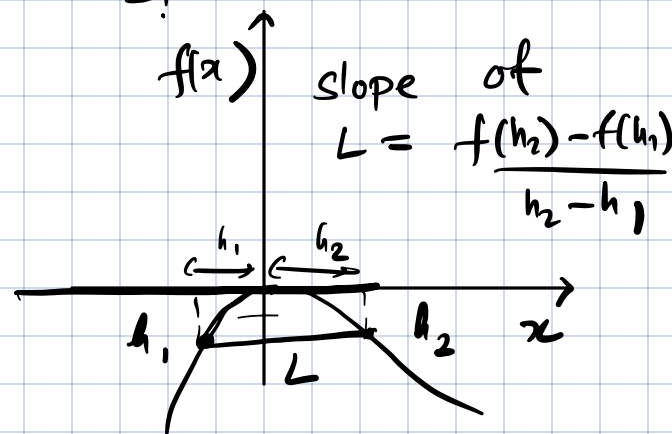
Exercise: Find n^{th} derivative of e^{-x^2} and $\tan^{-1} x$ at $x=0$.

$$f(x) = \underbrace{f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \dots}$$

$$f(x) \approx f(0) + f'(0)x$$

$$y = f(0) + f'(0)x$$

$$y = 0$$



Tool 15: $y = f(0) + f'(0)x$ is the tangent to the curve $y = f(x)$ at $x = 0$.

$$\begin{cases} f(x) \approx f(0) + f'(0)x & \text{for small } x. \\ f(x) \approx f(a) + f'(a)(x-a) & \text{for } x \text{ close to } a. \end{cases}$$

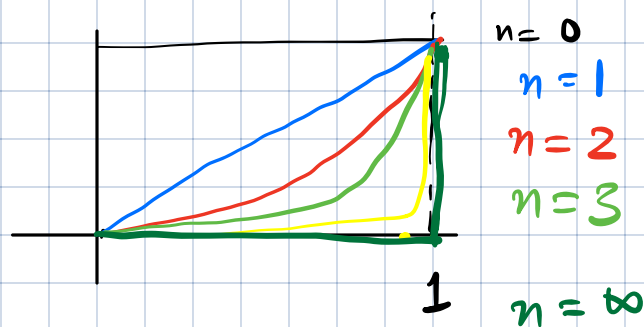
Linear approximation.

Convergence issues for functions:

Example 1 $f_n(x) = x^n$ for $0 \leq x \leq 1$

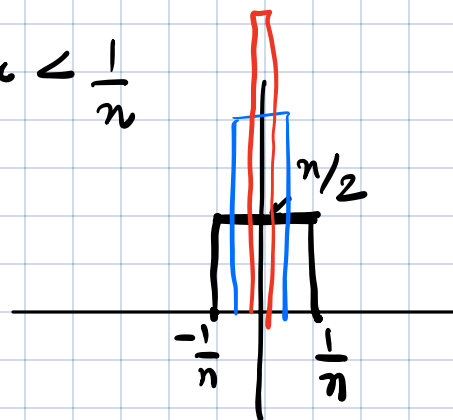
$$0 \leq x \leq 1$$

$$\lim_{n \rightarrow \infty} x^n = \begin{cases} 0 & \text{for } x < 1 \\ 1 & \text{for } x = 1 \end{cases}$$

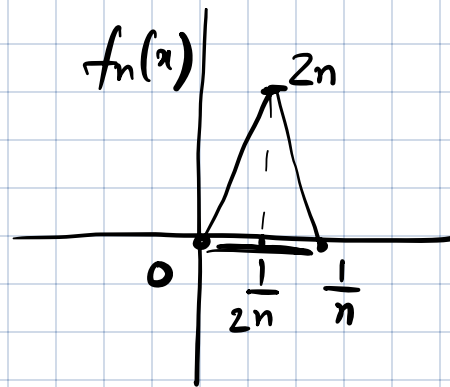


Limit of continuous functions may not be continuous.

Example 2: $f_n(x) = \begin{cases} \frac{n}{2} & \text{for } -\frac{1}{n} < x < \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$



$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0 & x \neq 0 \\ \infty & \text{for } x = 0 \end{cases}$$



Example 3:

$$\lim_{n \rightarrow \infty} \underbrace{f_n(x)} = f(x) = 0 \quad \text{for all } x.$$

$$\begin{aligned} 0 &= \int f(x) dx = \int \lim_{n \rightarrow \infty} \underbrace{f_n(x)} dx = \lim_{n \rightarrow \infty} \int f_n(x) dx \\ &= \lim_{n \rightarrow \infty} 1 \\ &= 1 \end{aligned}$$

Example 4: $f_n(x) = \frac{\sin nx}{n}$

$$|f_n(x)| \leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} f_n(x) = 0 = f(x)$$

$$f'_n(x) = \cos nx$$

(Uniform Convergence)

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

$$\boxed{\lim_{n \rightarrow \infty} f'_n(x) = f'(x)}$$

$$\frac{d}{dx} f(x) = \frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x)$$

$$\stackrel{?}{=} \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x)$$